

BST Physics Curriculum Vol 5 Chapter 9 — Identical Particles + Spin-Statistics v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

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Vol 5 Chapter 9 — Identical Particles + Spin-Statistics

Chapter motivation

Standard QM: identical bosons (integer spin) have symmetric multi-particle wavefunctions; identical fermions (half-integer spin) have antisymmetric wavefunctions (Pauli exclusion). Pauli 1940 + Streater-Wightman 1964 derive spin-statistics from relativistic QFT axioms (Lorentz invariance + microcausality + positive energy).

BST derives spin-statistics structurally from $\text{Pin}(2) \mathbb{Z}_2$ grading on Bergman $H^2(D_{IV}^5)$ — forced by rank = 2 per T1925 + T2443 RIGOROUSLY CLOSED Thursday. NO relativistic invariance assumption needed. Per Paper #133 v0.2 (Friday absorbed T2471).

Section 9.0b — Reader-grade 3-level pedagogy

Level 1: Pauli exclusion + boson/fermion partition follow from substrate $\text{Pin}(2) Z_2$ grading on Bergman $H^2(D_{IV}^5)$ per T1925 rank=2 + T2471 chirality $\gamma^5 = \text{Pin}(2) Z_2$ grading identification; no relativistic invariance assumption needed (BST contrast to Pauli 1940 / Streater-Wightman 1964).

Level 2 (graduate): T1925 rank=2 forcing Argument D establishes $\text{Pin}(2) Z_2$ grading on Bergman $H^2(D_{IV}^5)$; $\text{SO}(2)$ factor of $K = \text{SO}(5) \times \text{SO}(2)$ lifts via $\text{Pin}(2) =$ double cover of $\text{SO}(2)$. T2471 chirality $\gamma^5 = \exp(i\pi \cdot J_{\{\text{SO}(2)\}}^{\{\text{spinor}\}})$ IS the $\text{Pin}(2) Z_2$ grading object: $\gamma^{52} = I$; eigenvalues ± 1 ; γ^5 partition matches $\text{Pin}(2) Z_2$ partition exactly. Spin-statistics partition per Paper #133 v0.2: boson sector $V_-(p,q)$ with $q \in \mathbb{Z}$ (integer $\text{SO}(2)$ weight) $\rightarrow$ identity component of $\text{Pin}(2) \rightarrow$ symmetric multi-particle tensor products $\rightarrow$ Bose-Einstein statistics; fermion sector $V_-(p,q+1/2)$ with $q \in \mathbb{Z} + 1/2$ (half-integer $\text{SO}(2)$ weight via $\text{Pin}(2)$ double-cover) $\rightarrow$ spinor component $\rightarrow$ antisymmetric tensor products $\rightarrow$ Pauli exclusion + Fermi-Dirac statistics. The substrate-level proof requires NO Lorentz invariance + NO microcausality + NO positive-energy assumption (BST contrast to standard Pauli 1940 / Streater-Wightman 1964 axiomatic derivation). Standard QM identical-particle machinery (Slater determinant for fermions; symmetrization for bosons) inherits automatically. Multi-particle Fock space (Vol 1 Ch 7 dynamics) constructed on substrate K -type basis with $\text{Pin}(2)$ -graded symmetric/antisymmetric tensor product structure.

Level 3 (5th-grader): Quantum mechanics has a rule about identical particles: bosons (integer-spin particles like photons) can pile up in the same state; fermions (half-integer-spin particles like electrons) can't — this is Pauli exclusion, why electrons fill atomic orbitals one at a time, making chemistry possible. Standard physics derives this from relativistic QFT (a complex multi-step argument requiring Lorentz invariance). BST derives it from the substrate D_{IV}^5 's " $\text{Pin}(2) Z_2$ grading" — a binary structure forced by the BST integer rank = 2 (T1925 RIGOROUSLY CLOSED). The chirality operator γ^5 (proven Friday as T2471) IS this Z_2 grading. Bosons have $+1 \gamma^5$ eigenvalue; fermions have -1 . Pauli exclusion is automatic. BST doesn't need relativistic axioms — substrate geometry alone suffices.

Section 9.1 — Standard QM Identical Particles

Identical particles: symmetric tensor products for bosons; antisymmetric for fermions; Pauli exclusion for fermions. Slater determinant for fermion multi-particle states.

Section 9.2 — Standard Spin-Statistics Theorem

Pauli 1940 + Streater-Wightman 1964: relativistic QFT axioms (Lorentz invariance + microcausality + positive energy + non-trivial vacuum) $\rightarrow$ integer-spin Bose + half-integer Fermi.

Section 9.3 — BST Substrate $\text{Pin}(2)$ $\mathbb{Z}_2$ Grading

T1925 rank=2 Argument D: $\text{Pin}(2)$ $\mathbb{Z}_2$ grading on Bergman $H^2(D_{IV}^5)$. T2443 RIGOROUSLY CLOSED Thursday (Strong-Uniqueness C1).

$\text{SO}(2)$ factor of $K = \text{SO}(5) \times \text{SO}(2)$ lifts via $\text{Pin}(2) = \text{double cover of } \text{SO}(2)$; $\mathbb{Z}_2$ grading distinguishes integer (identity component) from half-integer (spinor component) $\text{SO}(2)$ weights.

Section 9.4 — T2471 Chirality $\gamma^5 = \text{Pin}(2)$ $\mathbb{Z}_2$ Grading

$\gamma^5 = \exp(i\pi \cdot J_{\{\text{SO}(2)\}^{\{\text{spinor}\}}})$ (T2471 Lyra Friday afternoon). $\gamma^{52} = I$; eigenvalues ± 1 ; γ^5 partition matches $\text{Pin}(2)$ $\mathbb{Z}_2$ partition exactly.

Section 9.5 — Paper #133 v0.2 Spin-Statistics

Per Paper #133 v0.2 (Friday absorbed T2471): - Boson sector $V_{\mathbb{Z}}(p, q)$ with $q \in \mathbb{Z} \rightarrow$ identity component of $\text{Pin}(2) \rightarrow$ symmetric tensor products - Fermion sector $V_{\mathbb{Z}}(p, q + 1/2)$ with $q \in \mathbb{Z} + 1/2 \rightarrow$ spinor component $\rightarrow$ antisymmetric tensor products

Boson/fermion partition $\equiv \gamma^5$ eigenspace partition $\equiv \text{Pin}(2)$ $\mathbb{Z}_2$ grading partition — three views of same substrate object.

Section 9.6 — Substrate-Level Proof: NO Relativistic Axioms

BST contrast to Pauli 1940 / Streater-Wightman 1964: substrate-level $\text{Pin}(2)$ $\mathbb{Z}_2$ grading argument requires NO Lorentz invariance + NO microcausality + NO positive-energy assumption. Standard QM machinery (Slater determinant, symmetrization) inherits from substrate.

Section 9.7 — Standard QM Recovery

Standard identical-particle machinery (Vol 5 standard QM curriculum + Vol 1 Ch 7 dynamics multi-particle Fock space) reproduces from substrate $\text{Pin}(2)$ $\mathbb{Z}_2$ grading.

Section 9.8 — Honest scope

- $\text{Pin}(2)$ $\mathbb{Z}_2$ grading + spin-statistics derived from substrate $\square$ (framework-grade)
- Standard relativistic-QFT spin-statistics axiomatic derivation recoverable from substrate
- Multi-particle Fock-space construction (Vol 1 Ch 7) cross-link
- Worked examples (Slater determinants, atomic orbital filling) pending

Section 9.9 — Connection to other chapters

- Vol 5 Ch 3 Angular Momentum + Spin (Pin(2) Z_2 grading framework)
- Vol 5 Ch 6 Hydrogen Atom (Pauli exclusion in orbital filling)
- Paper #133 v0.2 Spin-Statistics
- T1925 + T2443 + T2471 chain

— Lyra, Vol 5 Ch 9 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT